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Analysis of group ppt:

Analysis of the 1st group ppt:

Experiment	Observation	Inference
To the ppt add about 1mL of water and boil it	The ppt dissolves	Presence of lead
Test for Lead: i.) To one portion of the hot solution add about 1mL of K_2CrO_4 ii.) To an another portion of the hot solution add about 1mL of KI. To the yellow ppt add about 1mL of water, boil and cool.	A yellow ppt is obtained A yellow ppt is obtained. The yellow ppt dissolves on boiling , and on cooling golden spangles appear	

Reasoning:

Lead forms $PbCrO_4$ and PbI_2 ppt - Recrystallisation of lead iodide crystals appeared as golden yellow spangles.

Analysis of the 2nd group ppt:

To the ppt add about 1mL of dil HNO_3 and boil it.The ppt dissolves. Cool it.

i). To one portion of the solution add ammonium hydroxide	No ppt is obtained, but the solution is blue	Presence of copper
iii) Test for copper:To the blue coloured solution add about 1mL each of acetic acid and potassium ferrocyanide	A red brown ppt is obtained	Presence of copper

Reasoning

Prepare sodium stannite solution by mixing equal volume of about 1mL each of stannous chloride and sodium hydroxide.

With NH_4OH copper forms soluble $[Cu(NH_3)_4]^{2+}$ complex,

Copper forms a brown ppt of $K_2Cu[Fe(CN)_6]$

Analysis of the 3rd group ppt:

To the ppt add a pinch of sodium peroxide and boil it	A red or brown ppt is obtained A colourless solution is obtained	Presence of iron Presence of aluminium
i.) Test for iron:To one portion of the red ppt add about 1mL of dil HCl and boil it and then add about 1mL of potassium ferocyanide	A blue ppt is obtained	Presence of iron
ii.) To an another portion of the ppt add about 1mL of dil. HNO_3 boil it and then add about 1ml of KCNS	A blood red colouration is seen	Presence of iron
iii.) Test for aluminium:To the colourless solution add dil.HCl and shake it	A gelatinous white ppt is obtained	Presence of aluminium